

**BEFORE YOU READ — ANSWER FROM MEMORY**

1. A white patch or a red patch — which carries the **higher** risk of becoming cancer?
2. Which salivary gland has the **highest proportion** of malignant tumours?
3. Name the commonest **benign** and commonest **malignant** salivary gland tumour.

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Terminology — inflammation by site

Cheilitis	Lips	Glossitis	Tongue
Gingivitis	Gum	Stomatitis	Oral mucosa

2 White & red lesions — the discriminator

The definition does the work: leukoplakia is a white patch that **cannot be scraped off** and **cannot be explained by any other disease**. That single line excludes the two commonest impostors.

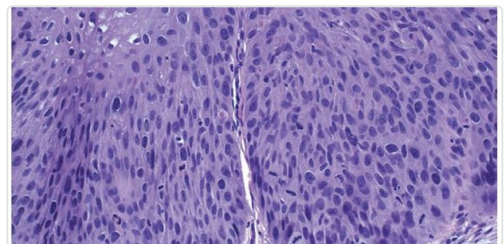
Lesion	Appearance	Scrapes off?	Malignant risk
Leukoplakia	White patch / plaque	No	3-25%
Erythroplakia	Red, velvety, flat or slightly depressed	No	>50%
Candidiasis	White patch	Yes	Not premalignant
Lichen planus	White patch	No — but is a defined disease	Not leukoplakia by definition



Leukoplakia — white plaque, buccal mucosa



Erythroplakia — red velvety area, far higher risk



Dysplasia: pleomorphism, loss of normal maturation

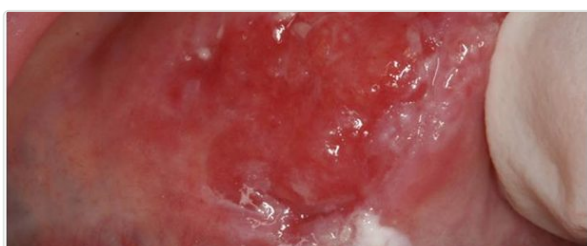
Every leukoplakia is **precancerous until histology proves otherwise**. Histology spans increased keratinisation without dysplasia → invasive keratinising SCC.

3 The progression to carcinoma

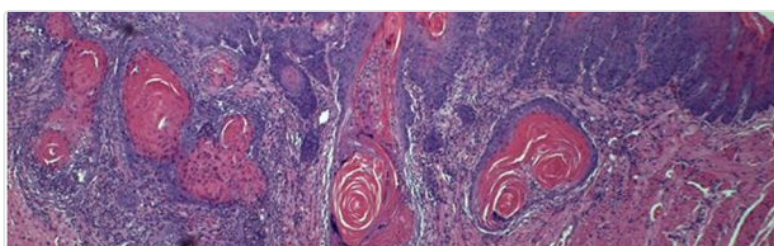
Risk factor	Strength of influence
Tobacco (pipe, smokeless especially)	Best-established influence
HPV 16 and 18	Found in 30-50% of cases; role in a subset
Alcohol	Weaker alone — but interacts with tobacco to greatly increase risk
Protracted irritation	Weakly associated

4 Oral squamous cell carcinoma — illness script

WHO gets it	Adults 40-70 , male 2:1 . Tobacco use is the dominant risk factor; alcohol multiplies it.
WHY it happens	Stepwise mucosal change: normal → leukoplakia/erythroplakia → dysplasia → invasive SCC . HPV drives a distinct subset.
WHAT you see	Ulceration and induration of oral mucosa. Histology: nests and islands of malignant keratinocytes invading stroma .
WHERE it spreads	Cervical lymph nodes first — most common regional site. Distant: mediastinal nodes, lungs, liver.
NEVER miss	95% of oral cavity cancers are SCC , and it is aggressive with poor survival. A non-healing oral ulcer needs biopsy, not reassurance.



Oral SCC — ulceration and induration



Nests of malignant keratinocytes invading stroma

5 Reactive & benign lesions — the one-liners

Lesion	Cause / trigger	The identifying feature
Recurrent herpetic stomatitis	Reactivated latent HSV-1	Groups of 1-3 mm vesicles → "cold sore"; resolves 7-10 days. Glassy intranuclear inclusions
Aphthous ulcer	Unknown; familial. Linked to coeliac, IBD, Behçet	Painful ulcer, fibrinopurulent exudate + erythematous halo ; 7-10 days
Fibroma	Chronic irritation → reactive CT hyperplasia	Submucosal nodule on the buccal bite line
Pyogenic granuloma	Minor trauma / irritation	Gingiva of children, young adults, pregnancy . Richly vascular, ulcerated; may regress
Mucocele	Duct blockage (retention) or rupture (extravasation)	Commonest lesion of minor salivary glands ; mucous-filled cyst



Orofacial herpetic infection



Aphthous ulcer with erythematous halo



Pyogenic granuloma on gingiva



Mucocele — mucous-filled cyst

Sialadenitis — inflammation of salivary gland. Causes: trauma; viral (**mumps**); bacterial (staph/strep, following **duct obstruction by stone** or dehydration); autoimmune (**Sjögren**).

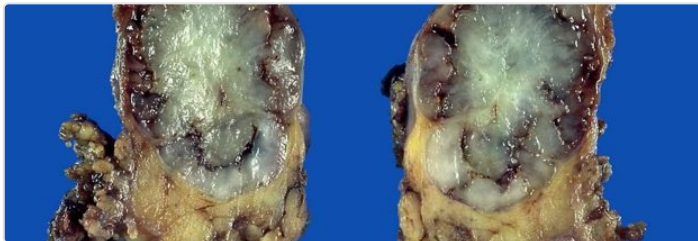
6 Salivary gland tumours — the size rule

The smaller the gland, the more likely the tumour in it is malignant. This one sentence generates the whole table — learn the rule, not the numbers.

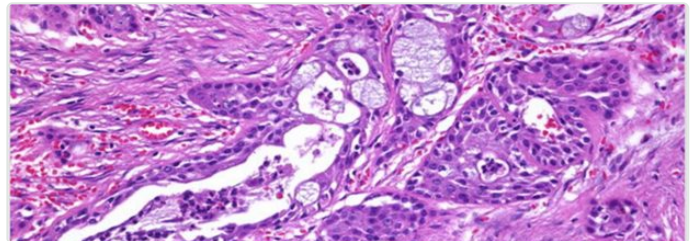
Gland	Share of all SG tumours	Proportion benign
Parotid (largest)	80%	80% benign
Submandibular	15%	50% benign
Sublingual / minor (smallest)	5%	40% benign — most likely malignant

Pleomorphic adenoma vs mucoepidermoid carcinoma

	Pleomorphic adenoma	Mucoepidermoid carcinoma
Rank	Commonest benign (50% of SG tumours; ~60% of parotid)	Commonest malignant (15%)
Also called	Benign mixed tumour	—
Who	Middle-aged females	—
Clinical	Painless, slow-growing, rounded, well demarcated	—
Composition	Ductal (epithelial) + myoepithelial cells → epithelial <i>and</i> mesenchymal differentiation; myxoid, hyaline, chondroid, even osseous matrix	Squamous + mucus-secreting + intermediate cells
Catch	Rekurs if incompletely excised ; may transform → carcinoma ex pleomorphic adenoma (malignant mixed tumour)	—



Pleomorphic adenoma — well demarcated, cut surface



Mucoepidermoid carcinoma — mixed cell populations

The full tumour list (Table 14-1)

Benign	Malignant
Pleomorphic adenoma (50%)	Mucoepidermoid carcinoma (15%)
Warthin tumour (5%)	Acinic cell carcinoma (6%)
Oncocytoma (2%)	Adenocarcinoma NOS (6%)
Cystadenoma (2%)	Adenoid cystic carcinoma (4%)
Basal cell adenoma (2%)	Malignant mixed tumour (3%)

Salivary gland neoplasms are only **2% of head and neck neoplasms**. The two in bold are the ones worth knowing in detail; the rest are worth recognising by name and rank.

7 Must-remember pearls

- **Red beats white for danger** — erythroplakia >50% malignant transformation vs 3–25% for leukoplakia.
- **Leukoplakia is a diagnosis of exclusion** — if it scrapes off it is candidiasis; if it is lichen planus, it is not leukoplakia.
- **Smaller gland = more likely malignant**. Parotid mostly benign; minor glands mostly malignant.
- **Pleomorphic adenoma recurs if incompletely excised** — the exam's favourite consequence of shelling it out.
- **Alcohol and tobacco interact** — combined risk far exceeds either alone.

8 Test yourself

1. A 58-year-old pipe smoker has a red, velvety, slightly depressed patch on the floor of mouth that cannot be scraped off. Compared with a white plaque at the same site, this lesion carries:

- A. A lower risk of malignant transformation
- B. An equal risk of malignant transformation
- C. A substantially higher risk of transformation
- D. No malignant potential

2. A 40-year-old woman has a painless, slow-growing, well-demarcated parotid mass. Histology shows ductal and myoepithelial cells within a chondroid matrix. The greatest risk of incomplete excision is:

- A. Facial nerve palsy
- B. Local recurrence
- C. Distant metastasis to lung
- D. Conversion to Warthin tumour

3. Which finding would exclude a diagnosis of leukoplakia in a patient with a white oral patch?

- A. The patient uses smokeless tobacco
- B. The patch is on the buccal mucosa
- C. The patch is wiped away with gauze
- D. Histology shows keratinisation without dysplasia

4. A tumour is discovered in a minor salivary gland of the palate. Compared with an identical-sized parotid tumour, it is:

- A. More likely to be malignant
- B. Less likely to be malignant
- C. Equally likely to be malignant
- D. Almost certainly a pleomorphic adenoma

5. A pregnant woman develops a richly vascular, ulcerated polypoid lesion on the gingiva. The most likely diagnosis is:

- A. Fibroma
- B. Mucocele
- C. Pyogenic granuloma
- D. Mucoepidermoid carcinoma

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 • C — Erythroplakia carries >50% risk of malignant transformation, far above leukoplakia's 3–25%.

2 • B — Pleomorphic adenoma characteristically recurs when incompletely excised.

3 • C — By definition leukoplakia cannot be scraped off; a patch that wipes away is candidiasis.

4 • A — The smaller the gland, the higher the proportion of malignant tumours (minor glands only 40% benign).

5 • C — Pyogenic granuloma; the gingival site in pregnancy gives it the name "pregnancy tumour".

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

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Figures reproduced from the lecture slides (Robbins).